Thread Pool

son: 5 · Concurrency · Category: Concurrency · Level: Intermediate · Build with W

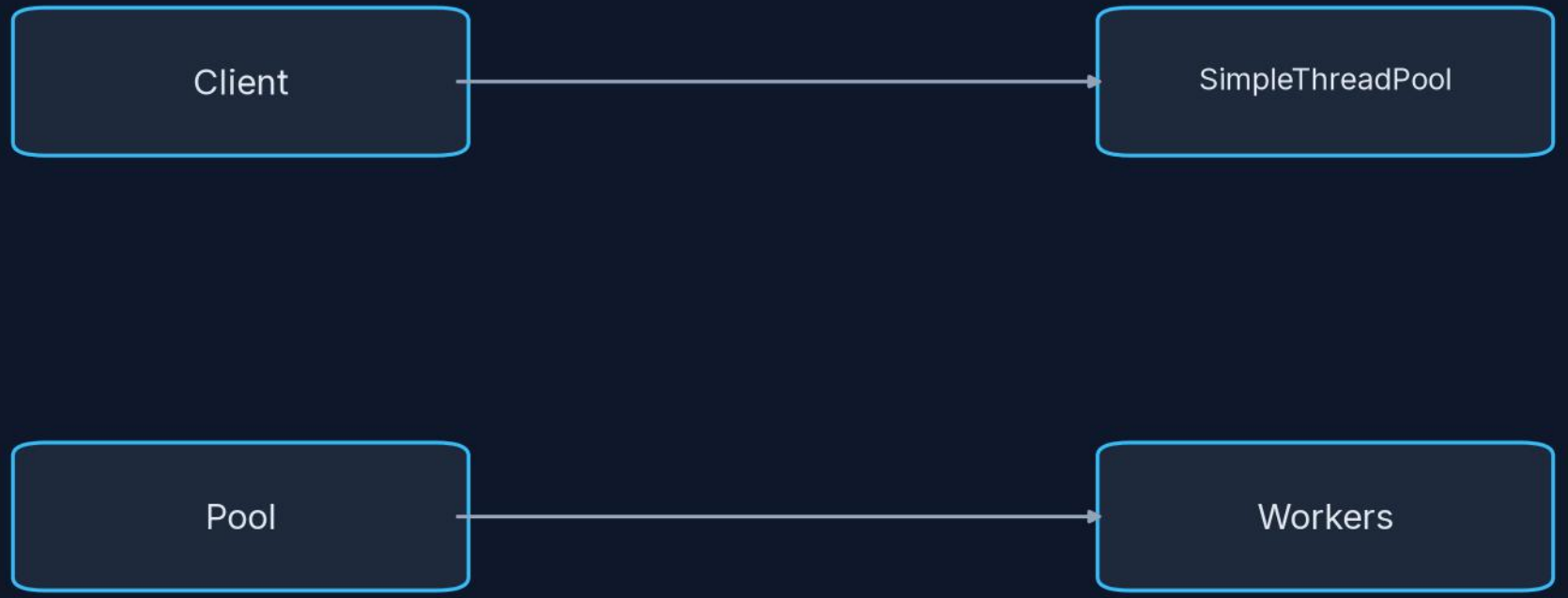
Today's Hook

- Today: Thread Pool as a StreamForge product force — not trivia.
- Runnable Demo on JDK 25 with captured stdout.
- Classic encoding → Java 25 / Zazr modernization → trade-offs.
- CapCut slides + full spoken script included.

The Problem

- Naive local fixes: more ifs, statics, nulls, or threads.
- Illegal states still compile; tests become order-dependent.
- Design reviews lack a shared name for the force.
- Interviews punish memorization without judgment.

Structure



Thread Pool structure

Real-World Scenario

- StreamForge creator platform is the running metaphor.
- Same names appear in naive code, refactor, and Demo output.
- Stay in one story for the full 10–12 minutes.
- Product language first; pattern vocabulary second.

Naive Code

```
// Naive sketch for Thread Pool
public class Naive {
  // duplicated logic / leaked details / unsafe sharing
  Object doIt(Object... args) {
    if (args == null) return null;
    return args[0];
  }
}
```

Why Naive Hurts

- Coupling spreads across controllers and infrastructure.
- Tests need sleeps or static reset hooks.
- New variants require shotgun edits.
- On-call cannot reason about failure modes.

Implementation

```java
// file: src/main/java/com/buildwithwaqar/patterns/concurrency/threadpool/SimpleThreadPool.java
package com.buildwithwaqar.patterns.concurrency.threadpool;

import java.util.concurrent.BlockingQueue;
import java.util.concurrent.LinkedBlockingQueue;
import java.util.concurrent.atomic.AtomicBoolean;

/** Educational pool — prefer Executors / virtual threads in real code. */
public final class SimpleThreadPool implements AutoCloseable {
    private final BlockingQueue<Runnable> queue = new LinkedBlockingQueue<>();
    private final Thread[] workers;
    private final AtomicBoolean open = new AtomicBoolean(true);

    public SimpleThreadPool(int size) {
        workers = new Thread[size];
        for (int i = 0; i < size; i++) {
            workers[i] = Thread.ofVirtual().name("worker-" + i).start(() -> {
                while (open.get() || !queue.isEmpty()) {
                    try {
                        var task = queue.poll(50, java.util.concurrent.TimeUnit.MILLISECONDS);
                        if (task != null) task.run();
                    } catch (InterruptedException e) {
                        Thread.currentThread().interrupt();
                        return;
                    }
```

Implementation (2)

```
                }
            });
        }
    }

    public void submit(Runnable task) {
        if (!open.get()) throw new IllegalStateException("closed");
        queue.add(task);
    }

    @Override
    public void close() {
        open.set(false);
        for (var w : workers) {
            try { w.join(1000); } catch (InterruptedException e) { Thread.currentThread().interrupt(); }
        }
    }
}
```

Demo.main

```
package com.buildwithwaqar.patterns.concurrency.virtualthreads;
public final class Demo {
  public static void main(String[] a) throws Exception {
    System.out.println("=== Virtual Threads walkthrough ===");
    System.out.println("fanOut=" + VirtualThreadDemo.fanOut(50));
  }
}
```

Walkthrough Output

```
=== Thread Pool walkthrough ===
completed=8
```

Step-by-Step Trace

- Each stdout line maps to a refactor stage.
- Re-run Demo locally on JDK 25 to verify.
- Differences usually mean a dirty tree — pull main.
- Pause the video and predict the next line before revealing.

Modern Java 25 / Zazr

```
// Java 25 / Zazr direction for Thread Pool
// records + sealed + switch
// Option/Either/Validation instead of null
// virtual threads for blocking fan-out
// inject interfaces at the edge
```

Trade-offs

- Clarity vs ceremony — do not spawn five types for two lines.
- Closed sealed worlds vs open SPI plugins.
- Teaching demos omit production idempotency/metrics.
- Prefer simpler language features when they erase the force.

Related Patterns

- Contrast with the neighbor pattern (force difference).
- Name the roles from memory in sixty seconds.
- Point to the season map for what comes next.
- Link the GitHub episode folder in the description.

Recap

- Force → naive pain → Thread Pool → Demo stdout.
- Modernize with Java 25 / Zazr when it shrinks code.
- Teach when NOT to use it.
- Practice: break a test, fix it, re-run Demo.